

RANKIN INLET, NU, Canada

WMO: 710830

Lat: 62.810N

Lon: 92.110W

Elev: 32

StdP: 100.94

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.0	(c) -39.5	(d) -44.4	(e) 0.0	(f) -40.9	(g) -43.2	(h) 0.1	(i) -39.4	(j) 18.8	(k) -26.3	(l) 17.3	(m) -28.6	(n) 5.8	(o) 350	(p) 1.092

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	19.9	14.0	17.5	12.5	15.5	11.4	14.4	19.0	13.0	16.9	11.7	15.0	5.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	8.9	16.0	10.7	8.0	14.6	9.7	7.5	13.1	40.5	19.1	36.8	17.0	33.5	15.1	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	14.0	12.6	-42.1	24.1	2.1	2.1	-43.6	25.7	-44.8	26.9	-46.0	28.1	-47.6	29.6
			-42.1	16.8	2.1	1.2	-43.6	17.7	-44.8	18.4	-46.0	19.0	-47.5	19.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.7	-29.9	-29.7	-24.4	-15.2	-5.1	4.7	11.1	10.1	4.4	-3.8	-15.9	-24.0
	DBStd	15.72	6.65	6.28	7.39	6.71	5.08	3.45	3.25	2.60	3.39	4.86	7.71	7.47
	HDD10.0	7279	1236	1111	1065	755	467	162	24	31	169	428	776	1056
	HDD18.3	10224	1494	1344	1323	1005	726	409	225	256	417	686	1026	1314
	CDD10.0	97	0	0	0	0	0	3	58	34	2	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	8	0	0	0	0	0	1	7	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.5	6.7	6.6	6.6	6.5	6.4	5.7	5.5	6.0	6.6	7.3	7.0	6.7
Precipitation	PrecAvg	320	11	10	13	19	17	23	43	51	49	40	25	17
	PrecMax	409	32	40	34	97	54	51	116	81	114	90	60	39
	PrecMin	219	2	0	2	1	0	1	4	11	13	10	6	4
	PrecStd	60	7	8	9	19	14	14	29	22	24	23	14	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.2	-10.4	-3.6	0.9	6.7	19.1	24.2	21.1	15.8	6.1	0.3	-3.4
		MCWB	-10.7	-10.5	-3.8	0.3	4.0	12.9	15.5	14.8	11.5	5.4	0.0	-4.0
	2%	DB	-13.6	-14.5	-7.7	-1.5	3.7	14.7	21.5	18.3	12.0	4.3	-1.2	-7.2
		MCWB	-14.1	-14.9	-8.0	-2.1	2.0	10.0	14.6	13.2	9.4	3.6	-1.7	-7.5
	5%	DB	-16.6	-16.7	-10.5	-3.4	2.1	12.3	19.0	16.4	10.2	3.2	-2.3	-10.1
		MCWB	-17.1	-17.1	-10.9	-4.0	0.8	8.4	13.3	12.2	8.2	2.5	-2.7	-10.4
	10%	DB	-19.2	-19.2	-13.1	-5.0	1.0	10.1	16.7	14.4	8.8	1.9	-3.7	-12.6
		MCWB	-19.6	-19.6	-13.5	-5.5	0.1	6.9	12.1	11.3	7.3	1.3	-4.2	-13.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.5	-10.7	-3.9	0.3	4.2	13.3	16.7	15.6	12.0	5.6	0.1	-3.9
		MCDB	-10.2	-10.5	-3.7	0.8	6.3	18.6	22.0	19.5	14.7	6.0	0.2	-3.4
	2%	WB	-14.1	-14.9	-8.1	-2.0	2.1	10.2	15.1	13.9	9.9	3.8	-1.5	-7.5
		MCDB	-13.6	-14.6	-7.7	-1.5	3.4	14.2	20.4	17.3	11.7	4.2	-1.2	-7.2
	5%	WB	-17.0	-17.1	-10.8	-3.9	1.1	8.5	13.7	12.6	8.6	2.5	-2.7	-10.4
		MCDB	-16.7	-16.8	-10.4	-3.5	1.9	12.0	18.4	15.5	9.8	3.0	-2.3	-10.0
	10%	WB	-19.6	-19.6	-13.4	-5.5	0.2	7.1	12.4	11.4	7.5	1.4	-4.2	-13.0
		MCDB	-19.2	-19.2	-13.0	-5.0	0.9	9.9	16.3	14.1	8.6	1.8	-3.7	-12.6
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.7	7.7	8.0	5.8	6.8	7.8	5.9	4.6	4.6	7.2	7.0
		MCDBR	9.2	9.2	9.6	7.5	5.7	11.3	11.9	9.4	6.9	4.5	7.2	9.3
		MCWBR	9.1	9.1	9.5	7.2	4.6	7.2	6.3	5.3	4.7	4.3	7.2	9.3
	5% WB	MCDBR	9.3	9.0	9.5	7.3	5.3	11.0	11.4	8.6	6.4	4.4	7.3	9.2
		MCWBR	9.1	9.0	9.5	7.1	4.4	7.3	6.6	5.4	4.6	4.4	7.3	9.2
Clear-Sky Solar Irradiance	taub		0.173	0.232	0.266	0.336	0.358	0.332	0.358	0.341	0.310	0.274	0.185	0.146
	taud		2.000	2.166	2.180	1.998	2.061	2.420	2.438	2.488	2.533	2.372	2.034	1.916
	Ebn at Noon		526	738	843	833	848	891	854	840	804	682	484	341
	Edh at Noon		31	62	91	138	142	102	98	85	67	51	29	16
All-Sky Solar Radiation	RadAvg		0.23	1.11	2.83	4.92	5.90	5.89	5.55	3.92	2.28	0.89	0.30	0.10
	RadStd		0.03	0.07	0.19	0.28	0.45	0.44	0.40	0.30	0.19	0.10	0.05	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page